HERIOT-WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

DEPARTMENT OF ACTUARIAL MATHEMATICS AND STATISTICS

F72ZA1
FINANCIAL MATHEMATICS 1

8 December 2005

Time allowed 2 hours

Attempt all **7** questions

Total Marks 60

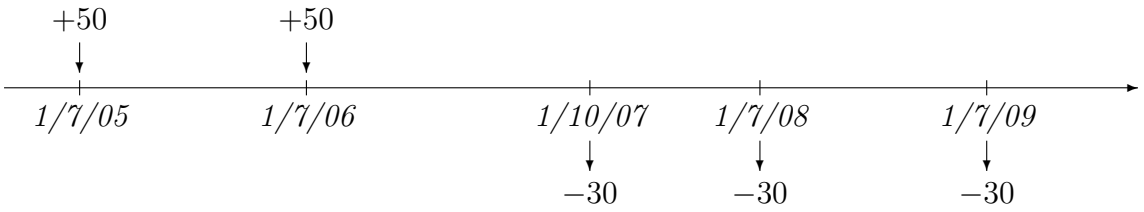
Only approved calculators may be used.

1. Fill in the missing values in the following table:

i	v	d	$i^{(4)}$	$d^{(2)}$
0.11	?	?	?	?

[4 marks]

2. The effective rate of interest will be 6% per annum during 2005, 3% per half-year during 2006 and 2007, and 1% per quarter-year thereafter. A series of cashflows is given as follows:



(a) What is the present value on 1 January 2005 of these cashflows?

(b) What is the accumulation on 1 January 2010 of these cashflows?

[7 marks]

3. Evaluate the following:

(a) $\ddot{s}_{\overline{20}|}^{0.06}$.

(b) $a_{\overline{\infty}|}^{0.115}$.

(c) $(Is)_{\overline{20}|}^{0.06}$.

[6 marks]

continued ...

4. (a) An annuity-certain is payable annually in advance for n years. The first payment is £1, thereafter payments increase at rate j per year. The effective rate of interest is i per year. Show from first principles that the present value at outset of this annuity is $\ddot{a}_{\overline{n}|}^{i'}$ where:

$$i' = \frac{1+i}{1+j} - 1.$$

[4 marks]

(b) An annuity-certain is payable annually in advance for 20 years. The first payment is £10,000, and subsequent payments increase at rate 0.03 (that is, by 3%) each year. During the first 10 years the effective rate of interest is 5% per year, thereafter it is 7% per year. What is the present value of this annuity at outset?

[8 marks]

[Total 12 marks]

5. (a) You borrow £100,000 to buy a house. The loan is to be repaid over 25 years by level annual repayments, payable in arrear. The interest rate is 6% per annum effective.

(i) What is the level annual repayment?

(ii) What is the capital outstanding just after the 8^{th} repayment has been made?

(iii) What is the interest content of the 9^{th} repayment?

[5 marks]

(b) Just after the 9^{th} repayment has been made, you ask the lender if you can miss the next three repayments. The lender agrees, but will charge a rate of interest of 9% per annum effective for the next five years, thereafter reverting to 6% per annum effective. What is the new annual repayment?

[5 marks]

[Total 10 marks]

continued ...

6. A company issues a bond with term 15 years, to be redeemed at par, with a coupon of 8% per annum payable half-yearly in arrear.

   (a) Some of the bond is purchased at outset by an investor who pays income tax at 25% and capital gains tax at 20%. The investor pays a price P per £100 nominal, such that their net redemption yield is 7% per annum effective. What is P?

   [7 marks]

   (b) Immediately after payment of the 20^{th} coupon, the investor sells the bond for exactly the same price P per £100 nominal as they paid for it. What net yield per annum does the investor achieve on the completed transaction?

   [3 marks]

   [Total 10 marks]

7. An investment fund has assets worth £100,000 on 31 December 2003. It receives an inflow of £20,000 on 1 January 2005 and pays an outflow of £50,000 on 1 January 2007. On 31 December 2007 it has assets of £89,825.

   (a) Find the money-weighted rate of return on the fund in the calendar years 2004–2007.

   [5 marks]

   (b) Suppose that the fund has assets of £133,560 on 31 December 2005. Find the linked internal rate of return during the calendar years 2004–2007, based on the subdivision of the period into 2004–2005 and 2006–2007.

   [5 marks]

   (c) Suppose also that the fund has assets of £102,000 on 31 December 2004 and £129,000 on 31 December 2006. Find the time weighted rate of return during the calendar years 2004–2007.

   [1 mark]

   [Total 11 marks]

Total 60 marks